Arista Access Node JAY-ACC-55 OSPF Routing Protocol Flap SOP

Node: JAY-ACC-55 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5774

Description

OSPF Routing Protocol Flap on the Arista Access Node JAY-ACC-55 at the Jayanagar Metro Site can occur due to the dynamic nature of OSPF's Shortest Path First (SPF) algorithm, which causes frequent route recalculations in response to changes in the network topology. These topology changes can be triggered by a variety of factors, including hardware or software failures, misconfigurations, or link flapping. In the context of the Arista access node, the OSPF protocol is responsible for exchanging routing information with neighboring routers and building a routing table that reflects the optimal paths to destinations on the network. However, when the OSPF adjacency with a neighboring router is lost, the access node will enter into the DOWN state, causing the routing table to be flushed and the SPF algorithm to be re-run, resulting in a flap of BGP routes. To prevent this flap, it is essential to identify and address the root cause of the OSPF adjacency loss and implement corrective measures to stabilize the network topology.

Pre-requisite Checks

- check: Verify the physical connection between JAY-ACC-55 and the neighboring router
- check: Check the OSPF neighbor state on JAY-ACC-55 using the 'show ip ospf neighbor' command
- check: Verify the OSPF network topology using the 'show ip ospf database' command

Potential Root Causes

Cause	Probability
Misconfigured OSPF interface	High
Hardware failure on JAY-ACC-55 or neighboring router	Medium
Software issue with OSPF implementation on JAY-ACC-55	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface 192.168.1.1 1 FULL/ - 00:00:38 10.0.0.1 Eth1/1 , Description: This command displays the OSPF neighbor state on JAY-ACC-55, which can help identify if the OSPF adjacency is lost or not.

Command: show ip ospf database, Expected Output: OSPF Router with ID (192.168.1.2) (Process ID 101) Type ID Adv Rtr Seq Age Interface Router 192.168.1.1 192.168.1.1 0x8000001d 00:00:07 Eth1/1 Net 10.0.0.0 192.168.1.1 0x8000001d 00:00:07 Eth1/1 , Description: This command displays the OSPF routing information on JAY-ACC-55, which can help identify if there are any issues with the OSPF network topology.
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```
Command: show bgp summary, Expected Output: BGP router identifier 192.168.1.2, local AS number 65001 BGP table version is 6 BGP - sent 0, received 10, withdrawn 0, suppressed 0, origin I Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd 192.168.1.1 4 65001 10 10 6 0 0 00:00:10 4/1 , Description: This command displays the BGP routing information on JAY-ACC-55, which can help identify if there are any issues with the BGP route churn.
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Resolution Steps

Step 1: Step: Verify OSPF interface configuration, Command: show run | inc ospf, Description: Verify that the OSPF interface is properly configured on JAY-ACC-55. Check for any misconfigured OSPF interface parameters such as network type, hello interval, dead interval, and priority.

Step 2: Step: Check for OSPF neighbor authentication, Command: show ip ospf neighbor authentication, Description: Check if OSPF neighbor authentication is enabled on JAY-ACC-55. If authentication is enabled, verify that the authentication keys are correctly configured on both JAY-ACC-55 and the neighboring router.

Step 3: Step: Restart OSPF process on JAY-ACC-55, Command: restart ospf, Description: Restart the OSPF process on JAY-ACC-55 to clear any OSPF-related issues. This will reset the OSPF database and restart the OSPF process.

Post-Check Verification

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Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface 192.168.1.1 1 FULL/ - 00:00:38 10.0.0.1 Eth1/1 , Description: Verify that the OSPF adjacency is stable and that the OSPF neighbor state is FULL.

Command: show ip ospf database, Expected Output: OSPF Router with ID (192.168.1.2) (Process ID 101) Type ID Adv Rtr Seq Age Interface Router 192.168.1.1 192.168.1.1 0x8000001d 00:00:07 Eth1/1 Net 10.0.0.0 192.168.1.1 0x8000001d 00:00:07 Eth1/1 , Description: Verify that the OSPF routing information on JAY-ACC-55 is stable and that there are no issues with the OSPF network topology.

Command: show bgp summary, Expected Output: BGP router identifier 192.168.1.2, local AS number 65001 BGP table version is 6 BGP - sent 0, received 10, withdrawn 0, suppressed 0, origin I Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd 192.168.1.1 4 65001 10 10 6 0 0 00:00:10 4/1 , Description: Verify that the BGP routing information on JAY-ACC-55 is stable and that there are no issues with the BGP route churn.
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Roll-back Procedures

- Step: Restart OSPF process on JAY-ACC-55, Command: restart ospf, Description: If the resolution steps do not resolve the issue, restart the OSPF process on JAY-ACC-55 to clear any OSPF-related issues. This will reset the OSPF database and restart the OSPF process.

- Step: Restore previous OSPF configuration, Command: configure replace /bootflash/ospf.conf, Description: If the restart of the OSPF process does not resolve the issue, restore the previous OSPF configuration on JAY-ACC-55 from the saved configuration file ospf.conf.